

n8n AI Travel Assistant

Project Explanation & Architectural Overview Report

Architecture: n8n Workflow + MCP Tooling

LLM Engine: Google Gemini (1.5 Flash/Pro)

1. Executive Summary & Core Intent

The **n8n AI Travel Assistant** is an automated, agentic workflow designed to generate personalized, real-time travel itineraries and automatically persist finalized trip data into external storage services including Google Docs and Google Sheets. Rather than relying on static parametric memory, the system dynamically orchestrates data fetching, user preference management, and document formatting.

2. Why Model Context Protocol (MCP) Was Necessary

Standard Large Language Models (LLMs) suffer from fixed knowledge cutoff dates and hallucinatory tendencies regarding real-time geographic data (e.g., specific hotel locations, opening hours, or dynamic pricing). Relying solely on the AI model's built-in memory limits itinerary accuracy.

Integrating **Model Context Protocol (MCP)** bridged the gap between generative reasoning and external real-time spatial services (via Overpass API / OpenStreetMap). MCP standardizes how the agent queries local coordinates, queries live geospatial features, and executes dynamic tool calling without needing hardcoded API middleware for each entity type.

3. MCP Tools Used & Functionalities

MCP Tool Name	Target Domain / Data Source	Core Function & Tool Execution Purpose
Hotel_Search_Tool	Overpass API (OSM)	Queries nearby hotels, resorts, and hostels using nwr nodes within specific geographical coordinates (e.g., 3km radius).
Attractions_Tool	Overpass API (OSM)	Fetches key cultural sites, historical landmarks, parks, and tourist destinations surrounding the coordinates.
Dining_Tool	Overpass API (OSM)	Locates local food spots, authentic restaurants, and cafes tailored to user culinary preferences.
Save_Travel_Preferences / Docs Tool	Google Docs & Sheets APIs	Executes document appending and record logging to persist the agent's finalized structured Markdown itinerary.

4. Core Benefits of Implementing MCP

- High Accuracy:** Eliminates hallucinated business names or outdated addresses by fetching raw OpenStreetMap spatial tags.
- Real-time Information Retrieval:** Obtains actual geographic boundaries, weather contexts, and current live entity nodes on demand.

- **Actionable Recommendations:** Provides genuine local transit points and real spot names rather than generic destination summaries.
- **Personalized Persistence:** Connects real-time queries with historic user preferences stored in Google Sheets to tailor budget and accommodation choices.

5. Technical Challenges & Architectural Solutions

Challenge 1: Overpass Connection Abort (HTTP 429 & Timeout Errors)

Problem: Public Overpass mirrors (e.g., kumi.systems) suffered server downtime or timed out when executing heavy node(around:5000) queries across large bounding boxes.

Solution: Switched endpoint to high-availability mirrors (maps.mail.ru / overpass-api.de), restricted radius to a balanced around:3000, used out center 10, and enforced a strict 30-second node execution timeout.

Challenge 2: Empty Array Returns for Non-Node Entities

Problem: Using strictly node queries omitted hotel buildings mapped as OSM ways or relations, yielding elements:[empty array].

Solution: Refactored queries to use nwr (Node, Way, Relation) with out center to reliably aggregate polygon-based hospitality structures.

Challenge 3: Multi-Tool Concurrent Execution Failure (MCP -32001)

Problem: Agent executed multiple MCP tools in parallel, triggering API rate limits.

Solution: Enforced sequential tool calling instructions in the AI Agent system prompt.